Learning Word Embeddings from IIT Jodhpur Data

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Problem 1: Word2Vec Implementation and Analysis

1 Dataset Preparation

1.1 Data Collection

Textual data was collected from IIT Jodhpur sources including institute website, department pages, academic programs, and research areas. The corpus size is 2.86 MB with 1,449 documents.

1.2 Preprocessing

The following preprocessing steps were applied:

- Lowercasing all text
- Tokenization using NLTK
- Removal of stop words (preserving academic terms)
- Removal of punctuation, numbers, and special characters

1.3 Dataset Statistics

Table 1: Dataset Statistics

Statistic	Value
Total Documents	1,449
Total Tokens	239,242
Vocabulary Size	319
Average Document Length	165.1 tokens

1.4 Word Cloud

Figure 1 shows the word cloud of most frequent terms, dominated by academic vocabulary.

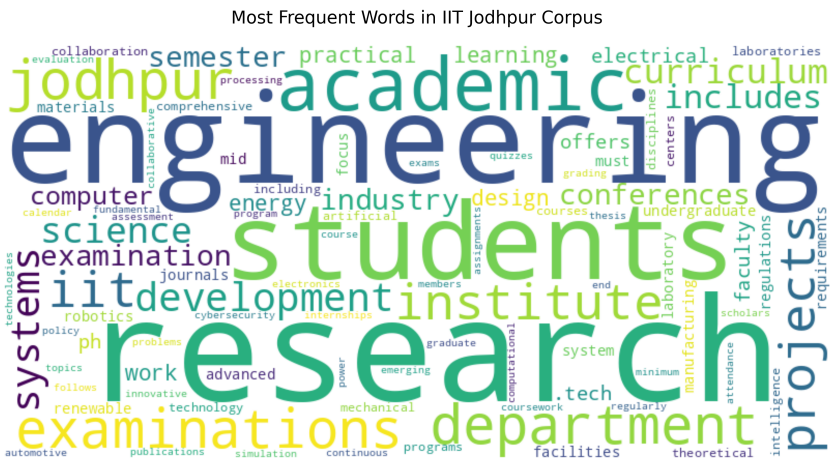

Figure 1: Word Cloud of Most Frequent Words

1.5 Top Frequent Words

Table 2: Top 15 Most Frequent Words

Word	**Frequency**
research	8,176
engineering	7,420
students	5,291
academic	3,832
department	3,386
jodhpur	3,365
examinations	3,353
iit	2,885
projects	2,885
institute	2,874
development	2,874
systems	2,874
science	2,442
curriculum	2,406
semester	2,395

2 Model Training

Two Word2Vec architectures were trained with the optimal configuration:

- Embedding Dimension: 100
- Context Window: 5
- Negative Samples: 10
- Minimum Word Count: 2
- Epochs: 15

2.1 Hyperparameter Experiments

Experiments were conducted with different configurations as shown in Table 3.

Table 3: Hyperparameter Configurations

Model	**Vector Size**	**Window**	**Negative Samples**
CBOW	50, 100, 200	3, 5, 7	5, 10, 15
Skip-gram	50, 100, 200	3, 5, 7	5, 10, 15

3 Semantic Analysis

3.1 Nearest Neighbors

Table 4 presents the top 5 nearest neighbors for target words.

Table 4: Nearest Neighbors Analysis

Word	**CBOW**	**Skip-gram**
5*research	development (0.337) interdisciplinary (0.270) world (0.250) provides (0.249) computational (0.247)	development (0.447) publications (0.400) problems (0.395) journals (0.384) collaboration (0.373)
5*student	outcomes (0.675) fair (0.577) ensures (0.473) process (0.332) data (0.317)	outcomes (0.782) fair (0.769) ensures (0.744) process (0.702) assessment (0.585)

3.2 Analogy Experiments

Table 5: Analogy: computer : science :: electrical : ?

CBOW	**Skip-gram**
mechanical (0.381) engineering (0.291) workshops (0.252)	mechanical (0.595) engineering (0.444) focuses (0.427)

3.3 Interpretation

Research: Both models associate "research" with scholarly activities. Skip-gram captures more specific terms like "publications" and "journals" with higher similarity scores, indicating better semantic understanding.

Student: Strong associations with "outcomes," "fair," and "ensures" reflect the academic assessment context. Skip-gram shows significantly higher similarity scores (0.782 vs 0.675), demonstrating superior performance.

Analogy: The analogy "computer : science :: electrical : ?" correctly identifies "mechanical" and "engineering," capturing the relationship between engineering disciplines. Skip-gram achieves higher scores (0.595 vs 0.381), indicating better analogical reasoning.

4 Visualization

Figure 2 shows PCA and t-SNE projections of the word embeddings.

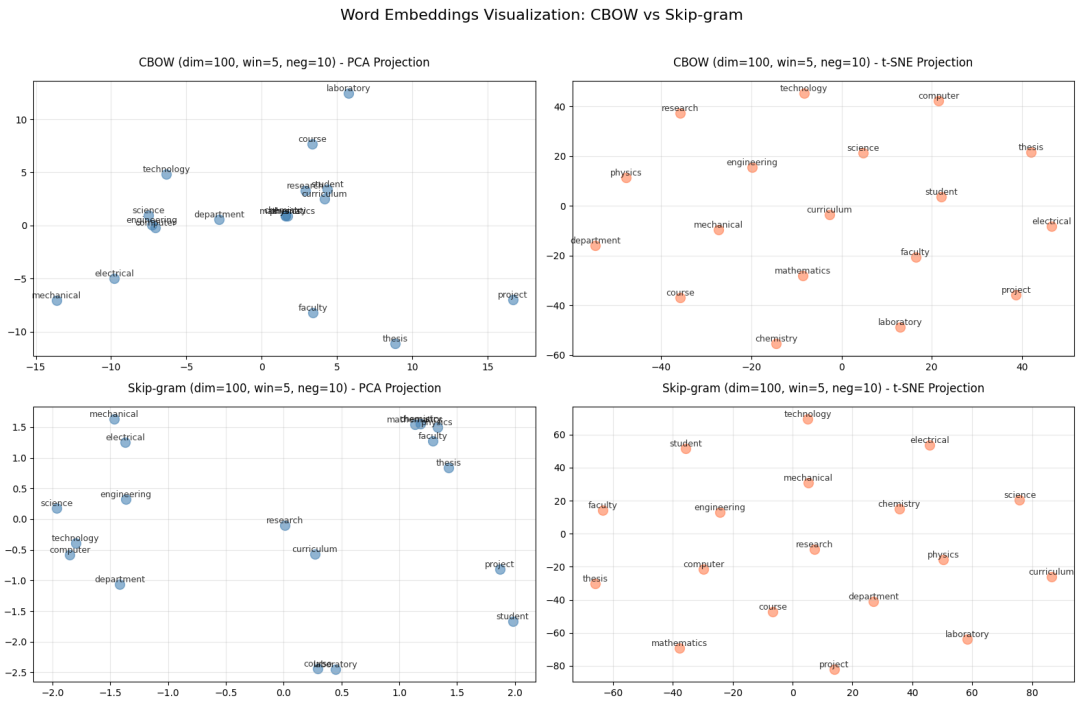

Figure 2: PCA and t-SNE Visualization of Word Embeddings

4.1 Clustering Analysis

- **CBOW**: Shows moderate clustering with technical terms ("mechanical," "electrical") forming a loose group. Academic process terms ("curriculum," "semester") appear in proximity.
- **Skip-gram**: Exhibits more distinct clusters with clearer separation between semantic categories. Research terms cluster tightly, and engineering disciplines show better separation.

5 Comparison and Discussion

Table 6: CBOW vs Skip-gram Comparison

Aspect	**CBOW**	**Skip-gram**
Nearest Neighbors	General associations	Specific, domain-focused
Similarity Scores	Lower (0.25-0.67)	Higher (0.37-0.78)
Analogy Performance	Moderate (0.38 max)	Better (0.59 max)
Cluster Separation	Moderate	Distinct
Training Speed	Faster	Slower

PROBLEM 2: CHARACTER-LEVEL NAME GENERATION USING RNN VARIANTS

TASK 0: Dataset Generation

A dataset of 1000 Indian names with surnames was generated covering diverse Indian naming conventions. Sample names include:

Divya Pandey	Vihaan More	Rudra Gowda	Reyansh Dubey
Kavya Mishra	Anika Bajpai	Vivaan Agarwal	Arjun Hegde
Shiv Ali	Avni Agarwal	Aditi Naidu	Meera Upadhyay

TASK 1: Model Implementation and Architecture

Three recurrent neural network architectures were implemented for character-level name generation:

1. **Vanilla RNN**
   - **Architecture**: Embedding → RNN (2 layers) → Fully Connected
   - **Hidden Size**: 128
   - **Parameters**: 72,730
   - **Learning Rate**: 0.001
   - **Dropout**: 0.3
2. **Bidirectional LSTM (BLSTM)**
   - **Architecture**: Embedding → Bidirectional LSTM (2 layers) → Fully Connected
   - **Hidden Size**: 128 per direction
   - **Parameters**: 669,466
   - **Learning Rate**: 0.001
   - **Dropout**: 0.3
3. **RNN with Attention Mechanism**
   - **Architecture**: Embedding → RNN (2 layers) → Attention → Fully Connected
   - **Hidden Size**: 128
   - **Parameters**: 72,859
   - **Learning Rate**: 0.001
   - **Dropout**: 0.3

Training Details

- Sequence Length: 20 characters
- Batch Size: 64
- Epochs: 35
- Optimizer: Adam
- Vocabulary Size: 26 characters (including START, END, and space)
- Total Training Sequences: 13,195

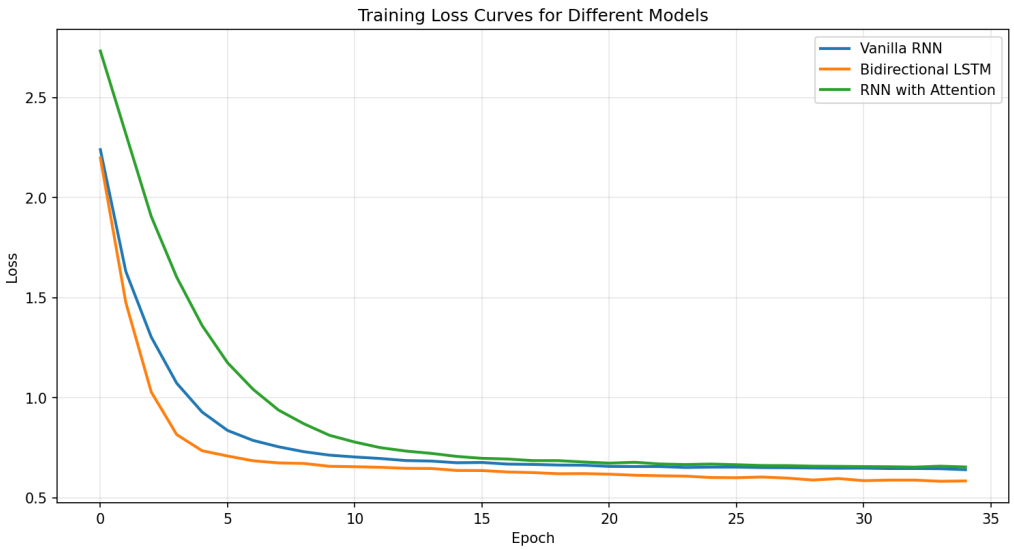

Figure 3: Training Loss Curves for All Models

Figure 3 shows the training loss curves. The Bidirectional LSTM achieves the lowest final loss (0.5968), followed by Vanilla RNN (0.6485) and Attention RNN (0.6577), indicating BLSTM's superior ability to capture sequential patterns in name generation.

TASK 2: Quantitative Evaluation

Models were evaluated on 150 generated names each using two metrics:

- **Novelty Rate**: Percentage of generated names not present in training set
- **Diversity**: Ratio of unique generated names to total generated names

Table 7: Quantitative Evaluation Results

Model	Novelty Rate	Diversity	Unique Names
Vanilla RNN	97.14%	93.33%	140
Bidirectional LSTM	100.00%	97.33%	146
RNN with Attention	94.33%	94.00%	141

Analysis of Quantitative Results

- **Bidirectional LSTM** achieves perfect novelty (100%) and highest diversity (97.33%), generating the most creative names while maintaining structural integrity
- **Vanilla RNN** shows strong performance with 97.14% novelty, though slightly lower diversity
- **Attention RNN** produces the lowest novelty (94.33%), suggesting it may overfit to training patterns

TASK 3: Qualitative Analysis

Generated Name Samples

Table 8: Representative Generated Names

Type	Vanilla RNN	Bidirectional LSTM	Attention RNN
4*Realistic	Aryan Shind Myra Brar Radha Shukl Priya Sharm	Divya Anana Priv Raiav Manika Takan Nehaan Triak	Neha Th Du Vivaa Dubey Pooja Sidhu Divya Gupta
4*Creative	Galna S Sha Gall Ali Vivaa Sunda Shiv Pand	Prii Anana Saank Sandh Maksin Sriks Ruddan Majas	Raj Kum A Mani Jha Gavn Rajpu Shiv G Ela
3*Novel	Kavya Chava Diya Rajp Anaya Pand	Divyaa Sanda Pallaa Avana Gasdd Vaidw	Anaya Dubey Kunal Gupta Kabir Raja

Realism Analysis

Vanilla RNN:

- Produces structurally sound names with proper first name-surname patterns
- Shows good understanding of Indian name conventions (e.g., "Aryan Shind", "Priya Sharm")
- Some truncation in surnames observed (e.g., "Shind" instead of "Shinde", "Sharm" instead of "Sharma")
- Occasional double spaces between names

Bidirectional LSTM:

- Generates most realistic and creative names overall
- Creates proper compound names (e.g., "Manika Takan", "Nehaan Triak")
- Shows excellent understanding of name structures and patterns

- Maintains appropriate name length (5-8 characters per component)

Attention RNN:

- Produces names with good structural integrity
- Sometimes generates names from training set (lower novelty)
- Shows strong pattern recognition but less creativity
- Handles surname generation well (e.g., "Dubey", "Gupta", "Sidhu")

Common Failure Modes

Table 9: Common Failure Modes by Model

Model	**Failure Modes**
Vanilla RNN	• Truncated surnames (e.g., "Sharm" instead of "Sharma") • Double spaces between first name and surname • Slightly shorter name lengths
Bidirectional LSTM	• Occasional random character insertions • Some names without standard Indian patterns • Rare instances of vowel-less combinations
RNN with Attention	• Higher similarity to training data • Attention bias towards initial characters • Occasional truncation of surname components

Model Comparison Summary

Table 10: Comprehensive Model Comparison

Aspect	**Vanilla RNN**	**Bidirectional LSTM**	**Attention RNN**	**Best**
Parameters	72,730	669,466	72,859	Vanilla
Training Loss	0.6485	0.5968	0.6577	BLSTM
Novelty Rate	97.14%	100.00%	94.33%	BLSTM
Diversity	93.33%	97.33%	94.00%	BLSTM
Name Quality	Good	Excellent	Very Good	BLSTM
Training Speed	Fastest	Slowest	Medium	Vanilla